ORF 309 – Homework 3, Q5a

Background: Expected Value

For a discrete random variable X, the expected value is a probability-weighted average over all possible values:

$$E[X] = \sum_x x \cdot P(X = x)$$

For example, for a fair die with outcomes $\{1, 2, 3, 4, 5, 6\}$ each with probability $\frac{1}{6}$:

$$E[X] = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + 3 \cdot \tfrac{1}{6} + 4 \cdot \tfrac{1}{6} + 5 \cdot \tfrac{1}{6} + 6 \cdot \tfrac{1}{6} = 3.5$$

For $X \sim \text{Binomial}(n, p)$, there is a closed-form shortcut:

$$E[X] = np$$

Q5(a): Probability the Plane is Overbooked

Setup. 202 tickets are sold. Each passenger independently fails to show up with probability 0.02, so each passenger shows up with probability $p = 0.98$. Let

$$X = \text{number of passengers who show up.}$$

Then $X \sim$ Binomial(202, 0.98).

The plane is overbooked when more than 200 passengers show up, i.e., $X > 200$. Since $X \in \{0, 1, \ldots, 202\}$, this means $X = 201$ or $X = 202$.

Calculation.

$$P(\text{overbooked}) = P(X = 201) + P(X = 202)$$

$$P(X = 201) = \binom{202}{201}(0.98)^{201}(0.02)^1 = 202 \cdot (0.98)^{201} \cdot 0.02$$

$$P(X = 202) = \binom{202}{202}(0.98)^{202}(0.02)^0 = (0.98)^{202}$$

Equivalent formulation. Letting $Y = 202 - X$ be the number of no-shows, we have $Y \sim \text{Binomial}(202,\, 0.02)$. Overbooking occurs when $Y < 2$:

$$\begin{aligned} P(\text{overbooked}) &= P(Y = 0) + P(Y = 1) \\ &= (0.98)^{202} + 202 \cdot (0.02) \cdot (0.98)^{201} \end{aligned}$$

Numerical answer.

$$\begin{aligned} (0.98)^{201} &\approx 0.017236 \\ P(X = 202) &= (0.98)^{202} \approx 0.016891 \\ P(X = 201) &= 202 \cdot 0.02 \cdot 0.017236 \approx 0.069634 \end{aligned}$$

$$\boxed{P(\text{overbooked}) \approx 0.016891 + 0.069634 \approx 0.0865}$$

Q5(b): Expected Ticket Revenue

The airline collects \$500 per *seated* passenger, and the plane seats at most 200. So the number of seated passengers is $\min(X, 200)$ and:

$$E[\text{revenue}] = 500 \cdot E[\min(X, 200)]$$

Computing $E[\min(X, 200)]$. Use the identity $\min(X, 200) = X - \max(X - 200,\, 0)$:

$$E[\min(X, 200)] = E[X] - E[\max(X - 200,\, 0)]$$

First term: By the Binomial shortcut,

$$E[X] = np = 202 \times 0.98 = 197.96$$

Second term: $\max(X - 200, 0)$ is nonzero only when $X > 200$. Since $X \le 202$:

$$\begin{aligned} E[\max(X - 200,\, 0)] &= 1 \cdot P(X = 201) + 2 \cdot P(X = 202) \\ &\approx 1(0.069634) + 2(0.016891) \\ &\approx 0.103416 \end{aligned}$$

Putting it together:

$$E[\min(X, 200)] = 197.96 - 0.103416 \approx 197.8566$$

$$\boxed{E[\text{revenue}] = 500 \times 197.8566 \approx \$98{,}928.29}$$